Q.1 (2 Marks Each)

1. **Define an Information Retrieval System and state any two objectives of such a system.**
2. **Differentiate between Information Retrieval and Data Retrieval on any four points.**
3. **List and briefly explain the components of an Information Retrieval System.**
4. **List the classes of classic Information Retrieval models.**
5. **Define Term Frequency and Inverse Document Frequency, and provide their respective formulas.**
6. **State any two limitations of the Boolean model.**
7. **What is user relevance feedback, and why is it used in query processing?**
8. **Explain keyword-based querying with a suitable example.**
9. **Define pattern matching in the context of query processing.**
10. Define a wildcard query and illustrate it with an example.

Q.2 (5 Marks Each)

11. **Explain the Information Retrieval process with the help of a neat block diagram, and discuss the role of each component.**
12. Explain the components, parts, and types of Information Systems in detail.
13. **Describe the working of search engines and explain how they differ from web browsers.**
14. Explain the Vector Space Model in detail. For the query `Q = "gold silver truck"` and the `documents D1 = "shipment of gold damaged in a fire"`, `D2 = "delivery of silver arrived in a silver truck"`, and `D3 = "shipment of gold arrived in a truck"`, compute the similarity of each document with the query using `TF-IDF` and cosine similarity, and rank the documents accordingly.
15. **Compare and contrast the Boolean, Vector Space, and Probabilistic Information Retrieval models on the basis of representation, matching function, ranking, and limitations. Also, draw the taxonomy of IR models.**
16. Explain the Fuzzy Set Model and the Extended Boolean Model with suitable examples.
17. Discuss the Probabilistic Model in detail, along with its ranking function.
18. Analyze the different types of query languages, including keyword-based queries, pattern matching queries, structural queries, and query protocols, with suitable examples for each.